WORKED SOLUTION

STEP III 2004, Question 10

Mechanics • Elastic Strings and SHM

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Read the question before you go on. It is in the STEP Support Programme's question database:

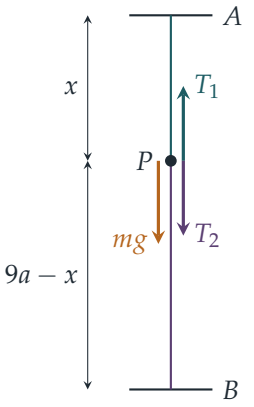
step.maths.org/questions/04-s3-q10

The key idea

Each string obeys Hooke's law only while it is taut, so the motion splits into stages according to which strings are taut. In each stage the acceleration is a constant multiple of $7a - x$, so the particle moves in simple harmonic motion about $x = 7a$, but with a different ω in each stage. We find the time and speed at the end of the first stage and use them to start the second.

Solution

We model P as a particle and the strings as light, obeying Hooke's law when stretched and exerting no force when slack. Take $x = AP$ measured downwards from A, so $PB = 9a - x$, and take downwards as positive.



The equations of motion

For $x > 6a$ the string AP is stretched by $x - 6a$, so its tension, pulling P up, is

$$T_1 = \frac{6mg(x-6a)}{6a} = \frac{mg(x-6a)}{a}.$$

The string BP is stretched by $(9a - x) - 2a = 7a - x$ when $x < 7a$, so its tension, pulling P down towards B, is

$$T_2 = \frac{6mg(7a-x)}{2a} = \frac{3mg(7a-x)}{a},$$

and $T_2 = 0$ for $x \geq 7a$, when BP is slack.

For $6a < x < 7a$, both strings are taut and Newton's second law downwards gives

$$\begin{aligned} m\ddot{x} &= mg - \frac{mg(x-6a)}{a} + \frac{3mg(7a-x)}{a} \\ &= \frac{mg}{a}(a - x + 6a + 21a - 3x) = \frac{mg}{a}(28a - 4x), \end{aligned}$$

so $\ddot{x} = \dfrac{4g}{a}(7a - x)$, as required.

For $7a < x < 9a$, only AP is taut, so

$$m\ddot{x} = mg - \frac{mg(x-6a)}{a} = \frac{mg}{a}(7a - x),$$

giving $\ddot{x} = \dfrac{g}{a}(7a - x)$, as required.

The time to reach B

Stage 1: $6a \leq x \leq 7a$. Put $y = x - 7a$. Then $\ddot{y} = -\omega_1^2 y$ with $\omega_1^2 = 4g/a$, which is simple harmonic motion about $x = 7a$. Starting from rest at $y = -a$ gives

$$x = 7a - a\cos\omega_1 t, \qquad \dot{x} = a\omega_1 \sin\omega_1 t.$$

Starting from rest at $x = 6a$, the particle accelerates downwards (since $\ddot{x} > 0$ there), so x increases from $6a$ and stays in this stage until $x = 7a$. That happens when $\cos\omega_1 t = 0$, first at $\omega_1 t = \pi/2$, so after time

$$t_1 = \frac{\pi}{2\omega_1} = \frac{\pi}{4}\sqrt{\frac{a}{g}},$$

with speed $a\omega_1 = 2\sqrt{ag}$, moving downwards.

Ask yourself The equation changes at $x = 7a$. How do we carry on?

The position and velocity are continuous at $x = 7a$, so the end of stage 1 gives the starting conditions for stage 2. We restart the clock there.

Stage 2: $7a \le x \le 9a$. Now $\ddot{y} = -\omega_2^2 y$ with $\omega_2^2 = g/a$, starting at the centre $y = 0$ with speed $2\sqrt{ag}$. So, with τ the time since $x = 7a$,

$$x = 7a + \frac{2\sqrt{ag}}{\omega_2} \sin \omega_2 \tau = 7a + 2a \sin \omega_2 \tau.$$

The amplitude is exactly $2a$, so the particle reaches $x = 9a$, which is the point B, when $\sin \omega_2 \tau = 1$, first at $\omega_2 \tau = \pi/2$. At that moment $\dot{x} = 2\sqrt{ag} \cos \omega_2 \tau = 0$, so it arrives at B with zero speed. This stage takes

$$t_2 = \frac{\pi}{2\omega_2} = \frac{\pi}{2}\sqrt{\frac{a}{g}}.$$

Hence the time taken for the particle to reach B is

$$t_1 + t_2 = \frac{\pi}{4}\sqrt{\frac{a}{g}} + \frac{\pi}{2}\sqrt{\frac{a}{g}} = \frac{3\pi}{4}\sqrt{\frac{a}{g}}.$$

Checks

Energy confirms that P just reaches B at rest. Take the start, with BP stretched by a, and the arrival at B, with AP stretched by $3a$. The elastic energy goes from $\frac{6mg\,a^2}{2 \cdot 2a} = \frac{3}{2}mga$ to $\frac{6mg\,(3a)^2}{2 \cdot 6a} = \frac{9}{2}mga$. The increase is $3mga$, exactly the gravitational energy lost in falling $3a$, so the kinetic energy at B is zero. Numerically, with $a = 0.7$ and $g = 9.8$, stepping the full equation of motion from rest at $x = 6a$ gives $x = 7a$ at $t \approx 0.2099$, the lowest point $x = 9a$ at $t \approx 0.6297$, and $\frac{3\pi}{4}\sqrt{a/g} = 0.6297$.

Common slips

- Using T_2 for $x > 7a$: a string can't push, so once BP is shorter than $2a$ it exerts no force.
- Using ω_1 for the whole motion, or forgetting to restart the second SHM with the speed $2\sqrt{ag}$ inherited from the first.
- Calling the second stage a quarter of a period without checking that the amplitude $2a$ really takes P to B.

Beyond STEP

This is a piecewise-linear oscillator: the restoring force is linear but the stiffness changes at $x = 7a$. Each half has its own phase-plane ellipse, and the trajectory is made of arcs of two different ellipses joined where the string goes slack. Systems like this, from bouncing balls to impact oscillators, are a standard starting point in nonlinear dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats oscillations and energy methods in depth.

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